Swin-MSP: A Shifted Windows Masked Spectral Pretraining Model for Hyperspectral Image Classification

Rui Tian, Danqing Liu, Yu Bai, *Member, IEEE*, Yu Jin, Guanliang Wan, and Yanhui Guo, *Member, IEEE*

***Abstract*— Deep learning has found widespread application in the hyperspectral image (HSI) classification, where transformer architectures based on self-attention have emerged as state-of-the-art (SOTA). The Swin-MAE framework utilizes a masked autoencoder approach with a shifted windows transformer as its backbone, demonstrating strong representational power and performance. This study proposes a shifted windows masking spectral pretraining (Swin-MSP) model, which achieves hierarchical modeling of hyperspectral data from local to global scales by introducing spectral masking pretraining techniques and a hierarchical architecture. To fit with this pretraining, we introduce the uniaxial continuous cross correlation layer (UC3L), a straightforward yet effective solution tailored for hyperspectral imagery masking. We design the shift frequency band transformer (SFBT) to hierarchically characterize spectral features. Experiments with publicly available datasets establish that our pretrained network significantly improves classification efficiency compared with SOTA networks. Furthermore, we systematically investigate the sensitivity of various datasets to pretraining hyper-parameters. The results underscore that the universal spectral representation acquired during the pretraining phase serves as a robust initialization for subsequent task-specific fine-tuning. It is noted that this work breaks from traditional vision transformer (ViT) approaches, offering a new perspective on hyperspectral dataset pretraining. The code is available at https://github.com/teaRRe/Swin-MSP.**

***Index Terms*— Hyperspectral image (HSI) classification, pretraining model, Swin-MAE, transformer.**

I. Introduction

HYPERSPECTRAL imagery represents an extension of multispectral remote sensing, offering higher spectral resolution and more abundant spectral information compared to traditional images. This abundant spectral data facilitates tasks like classification, detection, and analysis. Hyperspectral imagery has wide applications in domains like environmental monitoring [1], agriculture [2], geology [3], and medical diagnostics [4].

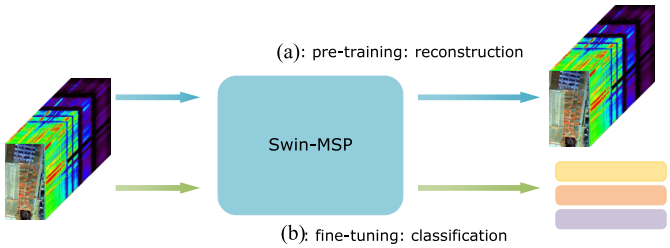


Fig. 1. (a) Specialized pretraining model for hyperspectral datasets, incorporating a shifted window masking spectral approach is proposed. (b) Perform downstream classification tasks with pretrained weights and biases.

In recent years, deep learning has become widely applied in hyperspectral image (HSI) classification. The emergence of transformer models featuring self-attention, popularized in natural language processing, has introduced new possibilities in computer vision (CV). Classification tasks using the vision transformer (ViT) [5] model have garnered significant attention, particularly for handling high-dimensional data such as HSIs. However, the quadratic complexity limits its self-attention mechanism. In response, the shifted window version of ViT, also known as Swin transformer [6], is proposed to address this limitation. This variation aims to reduce the computational and memory costs while retaining both the global receptive field and focused local features. Hyperspectral data often require manual annotation, which is both time-consuming and labor-intensive. Recently, unsupervised learning has garnered significant attention by learning the inherent structures or patterns from unlabeled data. In this paradigm, models are pretrained on a large amount of unlabeled data, and the learned intrinsic features are transferred to downstream fine-tuning tasks, providing a robust initialization for these tasks. In the field of visual pretraining, masking and reconstructing images is a viable method. This approach involves masking parts of the image and reconstructing the pixels, thereby uncovering the latent representations of the image through pretraining. MAE [7] has demonstrated the effectiveness of this method. However, the MAE, which uses a Transformer backbone, encounters similar challenges. To overcome image size limitations and optimize computation, Swin-MAE [8] has been proposed. This approach not only

Manuscript received 31 May 2024; revised 29 June 2024; accepted 15 July 2024. Date of publication 22 July 2024; date of current version 1 August 2024. This work was supported in part by the State Key Laboratory of Tibetan Intelligent Information Processing and Application, Tibetan Information Processing and Machine Translation Key Laboratory of Qinghai Province, under Grant 2023-Z-001; and in part by the Shandong Provincial Natural Science Foundation under Grant ZR2023MF110. *(Corresponding authors: Danqing Liu; Yanhui Guo.)*

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Digital Object Identifier 10.1109/TGRS.2024.3431517

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addresses the challenges of HSI data in unsupervised learning but also offers new directions and possibilities for research and practice in fields like HSI classification.

In the field of HSI classification, traditional ViT architectures suffer from quadratic computational complexity and high computational resource costs, while conventional MAE architectures have limited capability in handling high-dimensional data and have shown weaknesses in single proxy tasks. To address these issues, we propose a novel and versatile approach and establish a highly scalable backbone network. We introduce spectral masking to implement the pretraining process and demonstrate the effectiveness of the shifted window self-attention mechanism. The architecture is shown in Fig. 1.

The contributions of this article can be summarized as follows.

1) We propose the shifted windows masked spectral pretraining model, which achieves localized window partitioning along spectral channels to compute self-attention, utilizing the masking concept for pretraining.

2) To efficiently exploit the masking pretrain in the shifted window self-attention mechanism, we propose a straightforward yet effective uniaxial continuous cross correlation layer (UC3L) for hyperspectral imagery masking.

3) We introduce the shift frequency band transformer (SFBT), designed to hierarchically capture spectral features through pretraining and finetuning using distinct decoders.

4) We systematically evaluate the impact of pretrained hyperparameters on various datasets. This provides valuable insights into the critical role of the shifted window architecture in both pretraining and fine-tuning, thereby enhancing the efficacy and methodology of HSI classification models.

The remaining sections of this article are organized as follows. Section II reviews related work in the field of HSI classification. Section III provides a detailed description of the proposed shift window mask spectral pretraining (Swin-MSP) network, including how hyperspectral data is stratified for shift windows self-attention computation and the pretraining architecture. Section IV deploys extensive experiments and discussions to validate the effectiveness of the proposed approach and the roles played by each module. Finally, Section V draws the conclusion.

II. Related Works

A. Deep Learning Framework in HSI Classification

HSIs are acquired by measuring the spectral reflectance of the ground or other objects across multiple consecutive bands. Each pixel in the image contains rich spectral information. The primary objective of HSI classification is to categorize these pixels into distinct classes, which may represent various ground objects or materials.

In recent years, deep learning frameworks have been extensively explored for HSI classification, with the integration of various classic image models into this domain. Convolutional neural networks (CNNs) have been particularly prominent in these efforts, given their effectiveness in processing spatial information and their successful application in various image-processing tasks.

For example, in the work of [9], a combination of CNN and support vector machine (SVM) was proposed for HSI classification. Lee and Kwon [10] introduced a contextual deep CNN to enhance the contextual understanding of hyperspectral data. Yu et al. [11] presented considerable improvements to CNN architecture for HSI classification. Zhang et al. [12] proposed a CNN architecture tailored for HSI, emphasizing diverse region-based features. Yang et al. [13] conducted a comprehensive exploration of various CNN variants to discern their performance nuances. Vaddi and Manoharan [14] integrated spatial spectrum information with CNN to refine the classification process. Yu et al. [15] simplified 3D-CNN and introduced a new framework based on simplified 2-D–3-D CNN. In [16], a multibranch fusion network based on 2-D–3-D CNN for enhanced feature extraction was presented. In [17], an attention mechanism with feedback was introduced into CNN architecture to improve HSI classification performance.

However, traditional CNN architectures struggle to capture long-range dependencies between spectral bands, leading to limitations in handling global information. As a result, some work has proposed significant improvements. Wang et al. [18] proposed a multistage self-guided separation network for remote sensing scene classification. Tang et al. [19] introduced the object fine-grained change detection task and proposed a method based on frequency decoupling interaction. For subtle traits extraction, Zhang et al. [20] proposed a spatial-logical aggregation network with morphological transformation.

In cases of insufficient sample size, CNNs may suffer from overfitting or insufficient generalization. To address this issue, some studies have integrated generative adversarial networks (GANs) to enhance classification performance. Such as Zhu et al. [21] combined GAN and CNN to assess the effectiveness of GAN in HSI classification. Wang et al. [22] introduced the use of triple GANs for enhanced performance. In [23], a semi-supervised algorithm for GANs tailored for hyperspectral data training was proposed. Wang et al. [24] suggested an adaptive DropBlock enhanced GAN for improved HSI classification. Zhang et al. [25] presented a novel HSI classification method by amalgamating progressive growing GAN (PG-GAN) and Wasserstein GAN gradient penalty (WGAN-GP) concepts.

Some studies have introduced recurrent neural networks (RNNs) to address the challenge of long-range dependency handling, which CNNs may struggle with. In the pioneering work of [26], RNN was first employed for HSI classification. Subsequently, Zhang et al. [27] introduced a local spatial sequential method tailored for RNN in HSI classification. Paoletti et al. [28] proposed an innovative RNN classifier utilizing simple recurrent units for enhanced performance. Building on multiscale densely connected convolutional networks and an attention mechanism, Liang et al. [29] introduced a bidirectional RNN.

To address issues such as gradient vanishing and exploding gradients in RNNs, long–short-term memory (LSTM) networks have also been integrated for HSI classification. Tang et al. [30] proposed a two-branch feature extraction network composed of Gaussian pyramid multiscale transform and wavelet transform. Liu et al. [31] presented a novel deep

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learning framework employing bidirectional convolutional LSTM networks. Zhou et al. [32] integrated spatial spectrum information into the LSTM architecture. Mei et al. [33] introduced a spatial spectrum attention mechanism for LSTM networks, further enhancing their capabilities in HSI classification. These contributions collectively highlight the evolving and diversified approaches in RNNs for HSI classification.

B. ViT and Its Variants in HSI Classification

In the field of HSI classification, traditional deep learning models encounter challenges due to the distinctive properties of hyperspectral data. CNNs are limited by their restricted receptive fields, while RNNs and LSTM networks struggle with capturing long-range dependencies and achieving efficient parallel computation. These limitations have prompted researchers to seek alternative approaches. ViT models, which leverage self-attention as a fundamental mechanism, have emerged as a promising solution.

ViT transforms the traditional approach to CV by applying the Transformer model, originally designed for natural language processing, to CV. This involves converting image data into 1-D sequences as input to the transformer encoder. One of the ways to leverage ViT in HSI data necessitates the extraction of each pixel point along with its surrounding r-square block, while simultaneously extracting spectrum data. However, the implementation of ViT on HSI data faces two primary obstacles: the computational complexity of the attention mechanism on input features and the model's efficacy in extracting rich spectral information.

Some efforts have been made to improve the ViT architecture to enhance its efficiency on HSIs. For example, Zhao et al. [34] proposed a hyperspectral classification framework based on multiattention transformer and adaptive superpixel segmentation-based active learning. Bottleneck space-spectrum transform (BS2T) was proposed by Song et al. [35] to describe the long-term global dependency relationships among HS pixels. In [36], a new multibranch transformer structure was proposed to extract discriminative spectral features. Mei et al. [33] combined attention mechanism and LSTM to emphasize effective contextual information between spatial–spectral domains. A novel locally enhanced spectral transformer was presented in [37]. Mei et al. [38] proposed a hierarchical spectral transformer for HSI classification. Sun et al. [39] proposed a spectral–spatial feature tokenization transformer method to capture spectral–spatial features and high-level semantic features. To overcome the limitations of convolutional networks, Zhong et al. [40] proposed a new spectral–spatial transformer network. In [41], Hu et al. proposed an architecture with the global–local self-attention mechanism. A multilevel spectral–spatial transformer network was presented in [42]. To establish dependencies between long data sequences, Peng et al. [43] proposed a spatial–spectral transformer with cross attention. Hong et al. [44] rethink HS image classification and proposed a new backbone network SpectralFormer. Zhao et al. [45] proposed a novel unsupervised domain adaptation framework toward multilevel features and decision boundaries.

Traditional datasets may only contain HSI data, while real-world applications often involve multiple data sources. Some studies have begun to combine methods for handling multisource datasets to improve classification accuracy and robustness. Wang et al. [46] proposed a multistage information complementary fusion network based on flexible-mixup for hyperspectral-X image classification. Zhao et al. [47] proposed a fractional Fourier image transformer network, Wang et al. [48] proposed a method based on multiscale interactive information extraction for hyperspectral and synthetic aperture radar image classification. Gao et al. [49] combined adversarial complementary learning strategies to extract complementary information from multisource data. Gao et al. [50] proposed a spatial–spectral perception network to extract the advantages of different data sources and the cross information.

While ViT has shown promising performance on HSI classification tasks, its quadratic self-attention complexity poses a challenge for efficiently modeling large images. Thus, the Swin transformer employs a shifted window mechanism to confine attention computation to local windows, resulting in lower computational complexity and enhanced local feature modeling. Numerous studies have already begun to leverage these benefits. Huang et al. [51] proposed a 3-D Swin transformer that can adapt to the intricate 3-D properties of HSI classification. In [52], a new spectral Swin-transformer network was introduced. In [53], experts harnessed the potential of shifted windows in the spatial representations of both global and local features, presenting a dual self-attentional transform network.

C. Pretraining Model in HSI Classification

Most contemporary deep learning models depend on labeled datasets, however, labeling data manually is both tedious and challenging for some tasks, such as HSIs and medical images. Self-supervised learning is popular for its ability to avoid the cost of labeling massive datasets [54]. The primary objective of self-supervised learning is to enable models to learn efficient representations of unlabeled training data via the design of various pretraining tasks and loss functions, thereby promoting downstream tasks. Pretraining was initially applied in natural language processing as demonstrated by the BERT [55] model. which produced outstanding results in this area.

The MAE [7] model is the first to apply masked autoencoders in the visual domain. It randomly masks a portion of pixels in the input image, sending the remaining pixels (i.e., the unmasked portion) to an encoder to learn the image's latent representation. These representations are then fed into a lightweight decoder to reconstruct the full original image. The MAE model shows that rich image features can be learned through self-supervised learning, even with only partial image information. Recent research has shown promise in exploring the application of pretrained networks with masked autoencoder concepts in HSI classification. Kong et al. [56] proposed a scalable pretraining model based on the idea of MAE. Scheibenreif et al. [57] investigated the use of self-supervised masked image reconstruction to advance transformer models

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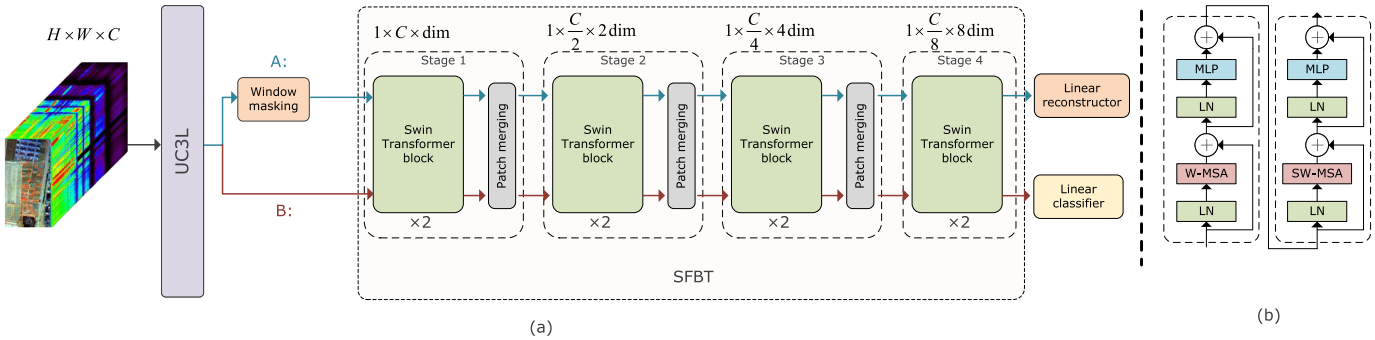


Fig. 2. (a) Architecture of our proposed Swin-MSP model consists of two main stages. Stage A involves the pretraining process, where the vectors extracted from the UC3L module enter the window masking module in a uniaxial direction and are then fed into the SFBT module for spectral feature reconstruction. Stage B comprises the fine-tuning and standard Swin transformer processes. The pretrained weights and biases will be utilized for fine-tuning, leading to land category classification through a linear classifier. (b) Two consecutive Swin transformer blocks will be applied.

for hyperspectral remote sensing imagery. Ibañez et al. [58] proposed a masked autoencoding spectral–spatial transformer with two distinct writing branches. Liu et al. [59] proposed a spectral masking self-supervised learning method with a cascaded encoder and decoder for deep semantic feature extraction. Mohamed et al. [60] proposed a decomposed spectral–spatial transformer incorporating a decomposed self-supervised pretraining process. Cao et al. [61] proposed a network combining contrastive learning and MAE principles.

Generally, pretraining is performed on large-scale datasets, making it challenging to pretrain on limited datasets. While compensating for a single proxy task using other methods (such as the masked image idea) has been widely studied, Swin-MAE [8] demonstrates that pretraining on a limited dataset is feasible on general images. However, techniques and concepts using shift-windows and masking autoencoders for HSI classification remain to be explored. To maximize the utility of existing publicly available small hyperspectral datasets, this study has created a spectral pretraining model with a shift window mask for generalized HSIs. The research systematically investigates the influence of diverse datasets on pretraining hyperparameters. The effectiveness of the proposed model is comprehensively evaluated through comparisons with networks lacking pretraining parameters. Our intent is to provide a novel backbone network and inspire further exploration in the realm of pretraining methodologies for HSI datasets.

III. Methodology

This section will introduce the details of the proposed model Swin-MSP. We will provide an overall structure first, followed by a detailed explanation of the UC3L and the SFBT. Subsequently, we will delve into the specifics of our pretraining scheme.

A. Overall Structure

The network architecture illustrated in Fig. 2. Incorporates key components for data flow in both pretraining and fine-tuning stages. The input data consists of each pixel and its neighboring pixels, as well as their spectral channels, forming a 3-D cube. In the UC3L module, this 3-D cube is projected as a 1-D vector with a length equal to the number of spectral channels, represented as $1 \times C \times \text{dim}$, where C represents the number of spectral channels. These feature vectors are fed into the SFBT module for window partitioning and self-attention computation.

In the SFBT module, we define a 1-D image with a fixed height of 1 and a width of C. During the multistage computation process, hierarchical feature extraction is achieved through PatchMerging applied solely to the spectral dimension. The patch merging in the first layer merges C spectral channels into $C/2$, and maps the output dimension to 2dim. This feature is then fed into the next stage's vectors. This process is repeated twice, resulting in $C/4$ and $C/8$ for stages 3 and 4, respectively, with feature sizes expanding to 4dim and 8dim. Stage 4 connects to a linear decoder, thus eliminating the need for patch merging.

The Swin transformer block we employ is consistent with the traditional Swin transformer block, comprising both a regular windows multihead self-attention module and a shift windows multihead self-attention module, as illustrated in Fig. 2(b). We will provide a detailed explanation in Section III-C.

B. Uniaxial Continuous Cross correlation Layer

In this study, a highly simple yet effective method is employed to embed HSI data into vectors that can be partitioned into uniaxial logical windows, facilitating spatial–spectral masking and hierarchical feature extraction. Specifically, for HSI data, each pixel corresponds to a land cover class, while latent relationships exist between a pixel and its surrounding pixels. To uncover these latent connections, we take the to-be-classified pixel and its surrounding pixels within a radius r together as one patch. For edge pixels, we use the filling zero method to meet the requirements. So the size of a single patch is $\text{patch_size} = h_{\text{patch}} = w_{\text{patch}} = 2r + 1$. Given HSI data $X \in \mathbb{R}^{H \times W \times C}$, We collect C spectral channels of the single patch together as a 3-D sample, so the size of each 3-D sample is $x' \in (p \times p \times C)$ (p represent patch size). As Fig. 3 shows, we flatten the samples and arrange them sequentially, resulting in a 2-D sample $x' \in (p \times p \cdot C)$. The spatial branch of the UC3L module performs a 2-D convolution on this sample

$$x_{\text{spa}} = \text{conv}_{\text{spa}}(x')$$

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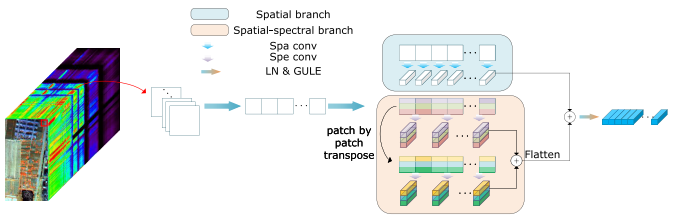


Fig. 3. Our UC3L architecture. The spatial branch will perform a 2-D convolution of the C uniaxialized patches of spectral dimension C. The spatial–spectral branch will perform a convolution with kernel and step size of $(1, p^2)$ on the uniaxialized vectors as well as their patch-by-patch transpose vectors, and the result will be flattened and added to the spatial branch.

where the kernel and stride of spatial convolution is (p, p), so $x_{\text{spa}} \in (1 \times C \times \text{dim})$. Meanwhile, the spatial–spectral branch performs three operations: convolution, transposed convolution, and addition

$$x_{\text{spe}} = \text{conv}_{\text{spe}}(x') + \text{conv}_{\text{spe}}(x'^{\text{T}})$$

where the kernel and stride of spatial–spectral convolution is $(1, p^2)$, x'^{T} is the patch-by-patch transposed representation of the original sample x', and $x_{\text{spe}} \in (1 \times p \times (C/p \times \text{dim}))$. We flatten x_{spe} to achieve multispectral feature fusion, so finally $x_{\text{spe}} \in (1 \times C \times \text{dim})$ and the results of the last two branches are added

$$l = \text{GELU}(\text{LN}(x_{\text{spa}} + x_{\text{spe}}))$$

where l is the feature vector fed into the SFBT module. This method fits the data into strips to efficiently utilize the shifted window mechanism. We will introduce the SFBT module in Section III-C.

C. Shift Frequency Band Transformer

1) Self-Attention in Local Uniaxial Windows: We propose establishing nonoverlapping logical windows along the spectral channel for local self-attention computation and shift windows to achieve cross-spectral connectivity. This approach eliminates the need for traditional ViT architectures to heavily rely on operations that reduce the dimensionality of the spectral channel. On a limited number of spectral channels, performing self-attention calculations can circumvent the secondary complexity arising from the traditional ViT architecture's operation around patch size.

We partition nonoverlapping logical windows along the uniaxial vector as depicted in Fig. 4(b). Specifically, assuming a input vector $1 \times C \times \text{dim}$ and window size of ws, the local self-attention computation is constrained to a $\text{ws} \times \text{dim}$ vector space. Meanwhile, the C spectral channels are divided into C/ws nonoverlapping windows. These windows will perform multihead attention computations at each layer. PatchMerging achieves spectral halving and feature doubling by segmenting the spectral dimension and combining the feature dimension. First, padding is applied on the H dimension to form $x \in (2 \times C \times \text{dim})$. Next, downsampling is performed along both the H and W dimensions, producing four vectors of shape $(1, C/2, \text{dim})$. These vectors are then concatenated along the C channel and subjected to a linear mapping, resulting in features of shape $(1 \times C/2 \times 2\text{dim})$.

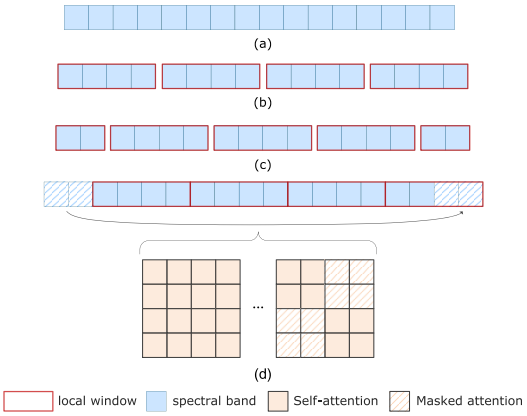


Fig. 4. Illustration of window partitioning, shifted windows, and self-attention masks. (a) Uniaxial vector from UC3L. (b) Windows partition. (c) Uniaxial shift. (d) Attention mask.

Each layer of the network will perform window-based self-attention computations and patch merging, in the final layer, the C channels will merge into $C/2^{l-1}$ and stop merging. We set ws to this value, which brings a specific characteristic. After passing through multiple layers, the windows are filled with the merged spectral, gradually capturing spectral information from local to global within the window.

2) Shifted Window Partitioning in Successive Blocks: The spectral information is continuous and context-dependent. To explore this potential information, we introduce uniaxial shift window connections. This is achieved by shifting windows on a 1-D vector and alternating their use between two consecutive Swin transformer blocks. As depicted in Fig. 4(c), the 1-D vector is divided into four windows of size $1 \times \text{ws}$, and window multi-heads self-attention (W-MSA) computation is performed. In the next block, the logical windows are shifted by ws/2 to perform shift-windows multiheads self-attention (SW-MSA) computation. The formula computed by two consecutive Swin transformer blocks is shown below

$$\begin{aligned}
\hat{z}^l &= W - \text{MSA}(\text{LN}(z^{l-1})) + z^{l-1} \\
z^l &= \text{MLP}(\text{LN}(\hat{z}^l)) + \hat{z}^l \\
\hat{z}^{l+1} &= \text{SW} - \text{MSA}(\text{LN}(z^l)) + z^l \\
z^{l+1} &= \text{MLP}(\text{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1}
\end{aligned}$$

where $\hat{z}^l$ represents the output of the self-attention calculation under a normal window partition, while $\hat{z}^{l+1}$ denotes the output under shifted window self-attention calculation. Both undergo layer normalization (LN), a multilayer perceptron (MLP), and residual connection sequentially. In each block, these processes are executed in order. z^{l-1} represents the output of the previous layer or, in the case of consecutive blocks, the output of the previous block. Meanwhile, z^{l+1} is passed on to the next layer or the next block in the continuous calculation sequence.

In 1-D shifting, the issue of connecting the vector's beginning and rear arises, which is meaningless in spectral channels. Therefore, it is necessary to constrain some attention computations within a specific window. The uniaxial constrain will only be applied to windows where the vector's beginning and rear are connected after being shifted. As depicted in

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Fig. 4(d). We mask out the meaningless attention to prevent it from interfering with the contextual information. Given that our pretraining proxy task is to reconstruct spectral features, and considering the influence of encoding on reconstruction, positional encoding has not been employed. This approach offers four advantages compared with traditional transformer block, as shown in Fig. 5.

1) Enhanced Modeling of Local Spectral Features: The limitation of attention within each window enhances the model's ability to capture and understand local spectral features effectively.

2) Comprehensive Learning of Interband Correlations: Shifting windows enable self-attention to cover neighboring spectral bands, facilitating comprehensive learning of interband correlations within the HSI.

3) Progressive Capture of Local to Global Features: Through patch merging, the approach gradually extends the capture range, progressively mastering local to global features in a systematic manner.

4) Improved Computational Efficiency: The utilization of uniaxial vectors avoids the quadratic complexity associated with standard self-attention mechanisms, resulting in increased computational efficiency during HSI processing.

D. Pretraining Scheme

Fig. 2 illustrates that the UC3L outputs spectral features in a uniaxial pattern during the pretraining process. Subsequently, we proceed to mask the spectra. Inspired by the Swin-MAE architecture, we designed a pretraining scheme tailored for HSI masks. As Fig. 6 shows, the scheme incorporates the following key designs.

1) Window Masking: In the classical Swin-MAE, the minimum masking unit is enhanced from a patch to a logical window composed of multiple patches. In the context of HSIs, we rethink the masking methodology. To preserve the continuity of interspectral information, we opt for patch-level stochastic masking over window masking. This approach encourages the model to learn potential features within each patch and the connection information between spectral bands.

2) Linear Reconstructor: Following the masked feature vectors' fusion by the blocks within the SFBT, the size and dimensionality undergo changes. To address this, we employ a linear reconstructor consisting of two linear layers. The linear rollback layer maps the features back to their original dimensions, while the linear decoding layer maps the dimensions to their original values. Subsequently, the reconstructor computes the mean absolute error against the original input to evaluate the quality of the reconstructed pixels. This simple linear reconstructor compels the encoder to focus its capabilities on modeling the mask tokens rather than leaving this task solely to the decoder.

IV. Experiments

In this section, we detail the experimental setup. First, we introduce the datasets and the experimental settings. Subsequently, we present and compare the results with those of the benchmark networks. Finally, we analyze the model and conduct ablation studies.

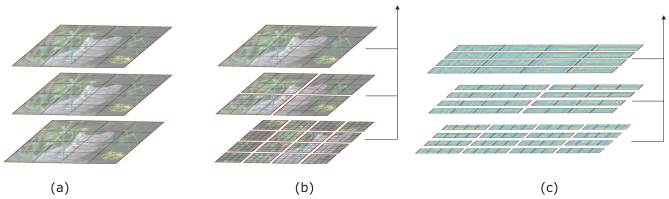


Fig. 5. (a) Original ViT architecture. (b) Swin-transformer's hierarchical partition window architecture. (c) Swin-MSP's uniaxial spectral window partitioning architecture.

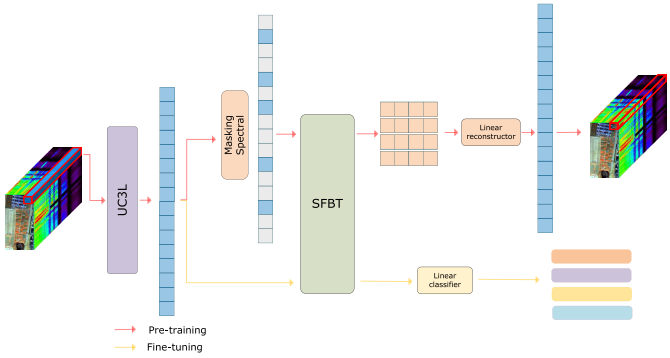


Fig. 6. Pretraining scheme. During the pretraining phase, the primary emphasis lies in feature reconstruction through the masking of spectral components, whereas fine-tuning involves classification training utilizing parameters acquired during the pretraining phase.

A. Dataset

1) Xuzhou (XZ): The XZ dataset [62] was collected by an airborne HYSPEX hyperspectral camera over the XZ peri-urban site in November 2014. The pixel resolution is 500 × 260, the number of spectral bands used in the experiment was 436; 68 877 pixels are used after removing background pixels. Ground truth annotations and class color representations are shown in Fig. 7(a), total samples and fine-tune samples are shown in Table I.

2) WHU_Hi_LongKou (LK): WHU-Hi dataset (Wuhan UAV-borne HSI [63]) is collected and shared by the RSIDEA research group of Wuhan University. The WHU-Hi-LongKou dataset was acquired from 13:49 to 14:37 on July 17, 2018, in Longkou Town, Hubei, China, with an 8-mm focal length headwall nano-hyperspec imaging sensor equipped on a DJI Matrice 600 Pro (DJI M600 Pro) UAV platform. The size of the imagery is 550 × 400 pixels, 204 542 pixels are used after removing background pixels. Ground truth annotations and class color representations are shown in Fig. 7(b), total samples and fine-tune samples are shown in Table II.

3) Pavia University (PU): This is the scene acquired by the ROSIS sensor during a flight campaign over Pavia in Northern Italy. The pixel resolution is 610 × 610 and contains 103 spectral bands. Using 42 776 pixels after removing unclassified background pixels, we use the first 100 bands. Ground truth annotations and class color representations are shown in Fig. 7(c), total samples and fine-tune samples are shown in Table III.

B. Experiment Setting

1) Network Setting: We quantitatively assess each model's classification performance using three widely used metrics: overall accuracy (OA), average accuracy (AA), and kappa

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(a) (b) (c)

Fig. 7. Ground truth annotations and class color representations. (a) XZ. (b) WHU_Hi_LongKou. (c) PU.

TABLE I
TOTAL AND FINE-TUNE SAMPLES FOR THE XZ DATASET

Class No.	Class Name	Total	Fine-tune
1	BARELAND1	26396	50
2	LAKES	4027	50
3	COALS	2783	50
4	CEMENT	5214	50
5	CROPS-1	13184	50
6	TRESS	2436	50
7	BARELAND2	6990	50
8	CROPS	4777	50
9	RED-TITLE	3070	50
Total		68877	450

TABLE II
TOTAL AND FINE-TUNE SAMPLES FOR THE LK DATASET

Class No.	Class Name	Total	Fine-tune
1	Corn	34511	50
2	Cotton	8374	50
3	Sesame	3031	50
4	Broad-leaf soybean	63212	50
5	Narrow-leaf soybean	4151	50
6	Rice	11854	50
7	Water	67056	50
8	Roads and houses	7124	50
9	Mixed weed	5229	50
Total		204542	450

TABLE III
TOTAL AND FINE-TUNE SAMPLES FOR THE PU DATASET

Class No.	Class Name	Total	Fine-tune
1	Asphalt	6631	50
2	Meadows	18649	50
3	Gravel	2099	50
4	Trees	3064	50
5	Painted metal sheets	1345	50
6	Bare Soil	5029	50
7	Bitumen	1330	50
8	Self-Blocking Bricks	3682	50
9	Shadows	947	50
Total		42776	450

coefficient (τ). The software platform consisted of Ubuntu 18.04, Python 3.8, and PyTorch 2.0. We utilized the AdamW optimizer with a cosine learning rate scheduler. The EPS and BETAS parameters for the AdamW optimizer were set to 1e-8 and (0.9, 0.999), respectively. During the pretraining phase, the base learning rate (BASE_LR) for the cosine scheduler was set to 5e-4, with a warmup learning rate (WARMUP_LR) of 5e-7. In the fine-tuning phase, the BASE_LR was adjusted to 1.25e-3, and the WARMUP_LR to 2.5e-7. Both phases employed a 10-epoch warmup period. In addition, the network parameters refer to the SimMIM [64] network.

During the training process, we conducted 500 epochs of training in the pretraining stage. In the fine-tuning stage, the network was initialized with pretrained weights and biases and trained for 100 epochs using 50 samples per class. It is noteworthy that we concentrate on studying and analyzing the reconstruction of spectral features in this work. Therefore, we set the HSI patch size to 5 and the feature mapping dimension to 96. For model depth, we used four layers for the LongKou and XZ datasets, and three layers for the PaviaU dataset, with each layer consisting of two Swin transformer blocks.

2) Comparison Networks: We assess the performance of the proposed Swin-MSP network in comparison to classical backbone networks and SOTA models. The networks under comparison include 1DCNN [65], 2DCNN [66], 3DCNN [67], as well as three SOTA models inspired by the ViT paradigm: SpectralFormer [44], MAEST [58], and MF [68]. These experimental results are derived from their optimal parameter configurations.

C. Quantitative Comparison

In the experimental results section of the proposed method, we present the performance of the proposed model in comparison to several benchmark models, using mean and standard deviation as statistical measures, as shown in Tables IV–VI. The proposed model consistently outperformed all compared models across various datasets. Specifically, the mean accuracy of our model achieves 94.08±1.55%, 96.99±0.90%,

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TABLE IV
COMPARISON WITH SOTA NETWORKS FOR XZ DATASET (%)

Class	1D CNN	2D CNN	3D CNN	SF (pixel)	SF (patch)	MAEST (pixel)	MAEST (patch)	morph Former	SwinMSP
1	60.89±23.23	83.05±2.57	71.54±7.17	84.35±3.98	90.40±3.05	86.36±3.19	91.20±2.91	**92.71±1.61**	92.67±4.35
2	76.83±15.34	85.04±8.64	84.68±8.37	97.34±0.52	94.90±1.58	97.00±0.50	94.69±1.87	97.78±0.79	**99.36±0.41**
3	59.44±12.19	69.01±7.24	61.81±5.85	74.97±11.39	35.41±13.71	91.42±3.99	95.47±2.31	92.92±4.19	**94.16±1.33**
4	50.74±24.12	69.01±7.24	77.93±2.87	82.47±7.21	**93.03±3.85**	90.77±2.91	91.23±3.54	90.03±4.00	93.00±2.06
5	70.60±14.15	68.04±11.91	84.17±2.97	74.88±6.08	89.16±4.98	88.34±4.36	93.45±2.66	**95.16±2.35**	94.05±1.93
6	33.95±16.70	79.79±4.87	40.69±5.56	61.74±6.11	89.27±3.67	74.18±5.64	91.50±4.79	93.16±6.01	**95.83±2.21**
7	45.29±15.20	53.37±17	37.81±8.81	68.51±11.20	93.07±12.52	87.18±3.18	92.84±6.26	**96.10±2.02**	93.55±5.98
8	55.93±27.85	59.83±13.65	80.25±10.08	94.64±5.05	97.03±1.48	97.53±0.90	97.35±1.98	93.88±2.11	**97.92±0.41**
9	46.35±25.66	88.46±3.44	75.29±3.08	64.11±26.43	95.92±2.48	94.68±3.10	**96.90±1.35**	95.99±3.42	95.28±3.67
OA	60.59±8.04	71.35±6.33	66.10±3.81	80.19±3.53	91.24±2.14	88.69±1.08	92.86±1.84	93.87±0.79	**94.08±1.55**
AA	59.4±15.74	79.73±8.26	66.10±3.98	78.11±4.49	91.99±1.48	89.72±1.06	93.85±1.10	94.19±0.80	**95.09±0.87**
Ka	53.27±7.71	65.78±7.32	60.17±4.00	75.31±4.28	89.03±2.62	85.87±1.29	91.04±2.26	92.26±0.99	**92.55±1.88**

TABLE V
COMPARISON WITH SOTA NETWORKS FOR LK DATASET (%)

Class	1D CNN	2D CNN	3D CNN	SF (pixel)	SF (patch)	MAEST (pixel)	MAEST (patch)	morph Former	SwinMSP
1	88.63±3.19	59.09±26.38	71.90±7.64	66.82±7.63	87.59±6.28	77.09±10.17	92.59±5.63	98.78±0.77	**99.18±0.15**
2	43.12±17.47	69.38±1.41	68.78±8.33	39.12±19.73	62.68±17.85	44.75±14.77	54.05±15.16	95.60±6.67	**98.33±1.14**
3	26.42±7.81	23.00±7.41	27.09±8.26	66.59±12.68	79.06±30.12	54.64±21.33	77.71±21.70	96.95±3.92	**98.92±0.73**
4	54.65±11.62	71.83±11.34	71.83±6.62	41.36±20.65	39.42±27.60	54.54±11.29	71.69±9.44	**94.14±5.56**	93.35±2.93
5	19.05±12.77	36.82±19.73	36.96±6.40	52.09±25.20	63.31±22.36	58.18±20.46	71.09±18.25	95.23±18.08	**99.15±1.09**
6	83.82±8.36	56.93±17.67	66.05±14.72	89.13±4.69	90.61±5.76	93.38±4.50	94.31±3.69	96.10±14.36	**98.88±0.34**
7	95.16±4.24	97.50±0.83	97.51±0.49	99.97±0.02	99.77±0.15	**99.91±0.05**	99.80±0.13	99.49±0.08	99.03±0.57
8	66.10±10.24	76.73±5.05	80.12±3.63	87.27±2.90	82.88±11.69	84.68±3.87	86.48±3.62	90.72±84.55	**93.02±1.80**
9	41.57±18.81	88.12±2.82	83.36±4.91	37.99±5.00	78.94±4.73	39.77±7.00	78.15±5.82	88.88±29.49	**96.58±0.64**
OA	72.77±3.70	71.99±8.92	74.59±4.30	69.66±6.18	74.85±8.86	75.83±3.69	85.79±2.07	96.65±0.46	**96.99±0.90**
AA	56.11±3.46	74.95±10.43	78.18±8.26	64.48±2.23	76.03±2.92	67.44±2.50	80.65±3.25	93.71±8.95	**97.38±0.42**
Ka	66.17±4.41	65.58±10.08	68.42±5.09	63.03±6.62	69.49±10.03	69.97±4.20	81.93±2.48	95.63±0.77	**96.08±1.15**

TABLE VI
COMPARISON WITH SOTA NETWORKS FOR PU DATASET (%)

Class	1D CNN	2D CNN	3D CNN	SF (pixel)	SF (patch)	MAEST (pixel)	MAEST (patch)	morph Former	SwinMSP
1	57.00±16.95	66.57±3.64	71.90±7.64	64.65±7.25	61.32±6.42	57.31±10.07	72.88±6.54	87.85±6.72	**94.73±1.48**
2	62.59±20.81	61.92±6.54	82.09±3.72	54.57±12.72	62.57±9.44	55.15±11.53	66.42±10.13	91.90±5.48	**97.33±0.86**
3	34.78±19.03	63.57±6.68	67.12±5.49	27.96±11.57	53.49±11.65	36.04±14.26	82.17±9.06	83.31±12.05	**93.01±4.51**
4	63.54±13.24	81.13±4.39	94.36±1.78	**97.80±0.88**	96.17±1.68	97.41±1.03	95.54±3.02	91.37±4.15	97.75±0.96
5	76.49±14.77	98.52±1.11	97.22±1.75	99.22±0.11	99.99±0.02	99.22±0.27	99.95±0.12	99.92±0.13	**100.00±0.00**
6	38.79±13.83	64.10±5.06	74.00±6.31	50.95±12.73	73.52±13.25	57.64±13.01	78.43±9.27	73.05±21.62	**97.96±0.95**
7	48.50±5.21	55.43±5.48	74.18±4.98	82.48±15.08	88.31±3.81	94.31±2.66	88.11±5.29	93.63±4.83	**99.26±0.83**
8	59.64±13.50	78.94±12.21	84.36±2.99	72.87±10.23	78.16±7.59	72.25±12.16	69.59±12.68	91.22±5.91	**95.13±3.09**
9	76.77±16.90	87.70±4.71	97.85±1.38	99.91±0.10	99.86±0.12	99.93±0.05	99.80±0.17	98.07±1.03	**99.99±0.03**
OA	58.39±8.98	64.97±3.86	79.31±3.25	62.44±5.34	69.67±3.24	62.81±5.47	74.34±3.46	88.97±3.84	**96.83±0.38**
AA	43.21±4.78	74.71±5.51	88.08±4.34	72.27±2.07	79.27±1.71	74.36±1.93	83.66±1.05	90.04±3.71	**97.24±0.42**
Ka	49.75±8.82	58.72±4.06	73.97±3.86	53.47±5.07	62.28±3.35	54.36±5.43	67.86±3.62	85.36±5.20	**95.80±0.50**

and 96.83±0.38% in XZ, LK, and PU datasets, respectively. This demonstrates not only superior performance but also the competitive stability of the proposed method, as indicated by lower standard deviations in our results. These findings underscore the effectiveness of the proposed model in handling HSI classification, establishing its potential as a robust tool in the field. Figs. 8–10 depict the classification maps from single-experiment evaluations comparing the proposed network with SOTA networks. These images visually represent the output of each network when applied to the same dataset under identical experimental conditions. The classification maps are color-coded to indicate different classes, allowing for a direct visual assessment of each network's classification accuracy and effectiveness.

D. Model Analysis

In this section, we explore the impact of initializing the proposed model with pretrained weights and biases compared to a random initialization without pretraining on classification performance. We also conducted a comprehensive study on the effect of pretraining epochs across datasets of varying quality.

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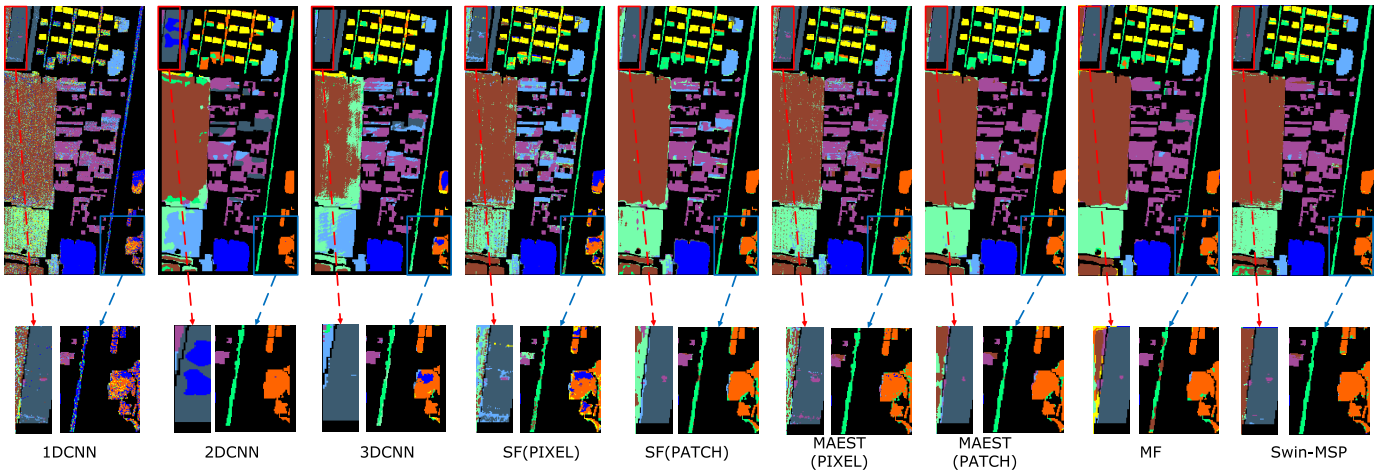


Fig. 8. Classification maps and typical regions of the XZ dataset generated using SOTA networks.

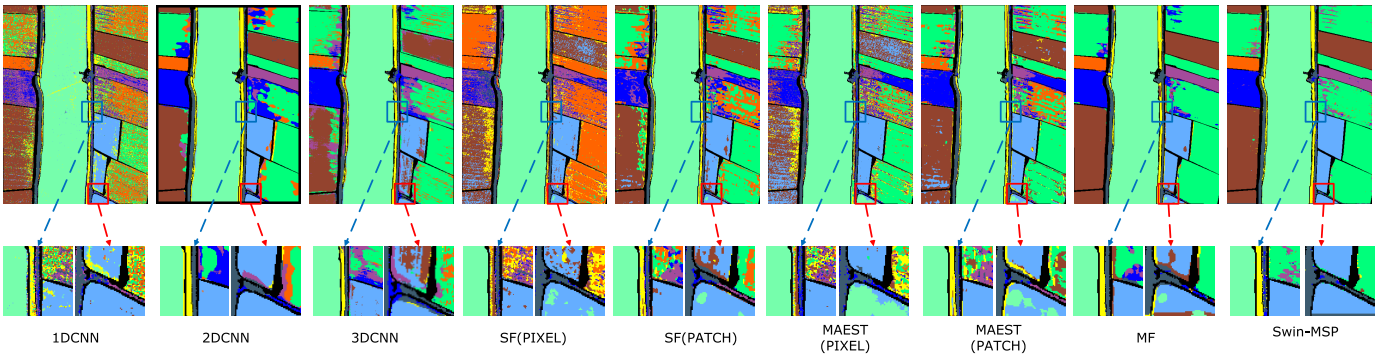


Fig. 9. Classification maps and typical regions of the LK dataset generated using SOTA networks.

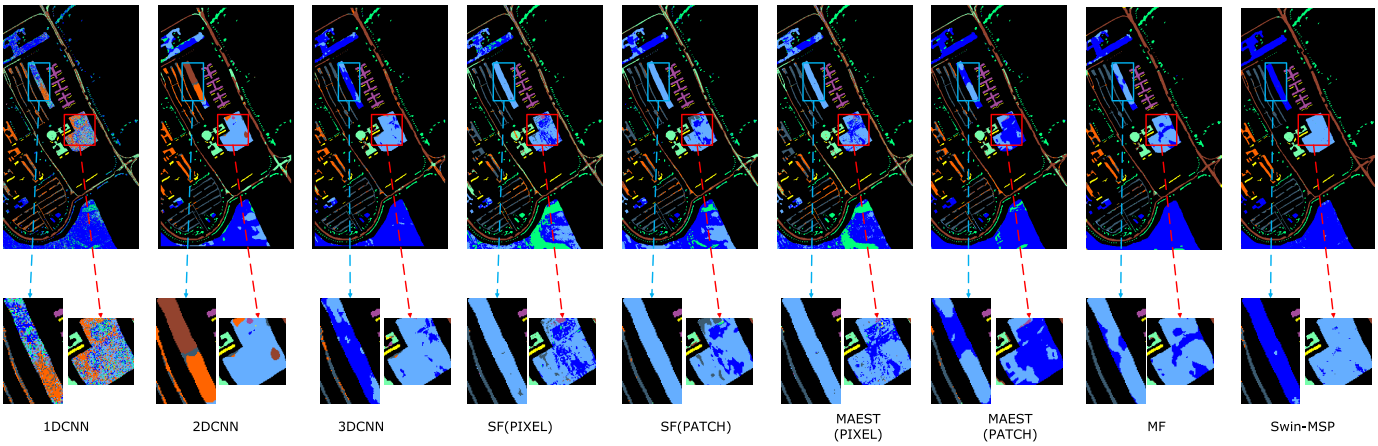


Fig. 10. Classification maps and typical regions of the XZ dataset generated using SOTA networks.

Our findings reveal that models initialized with pretrained weights consistently outperform those with random initialization across all tested datasets. Moreover, we observed a significant reduction in training time for models equipped with pretrained weights, underscoring that our method enhances both performance and efficiency. This underscores the wide applicability of our pretraining approach to diverse datasets.

1) Effect of Pretrain: Fig. 11 shows the performance of the proposed network. We performed a comparative analysis between the baseline performance, obtained from training from scratch, and the fine-tuned performance after pretraining, using the OA from a single training session as the performance metric. Under identical parameters and training iterations, the fine-tuning with pretrained weights and biases substantially outperformed conventional training methods. This demonstrates that the proposed model is capable of acquiring a significant amount of latent representations through masked spectral pretraining, significantly enhancing its efficiency in subsequent classification tasks.

2) Effect of Training Quality: In Fig. 12, we demonstrate the fine-tuning performance of our model across various pretraining epochs. The line graphs represent three performance

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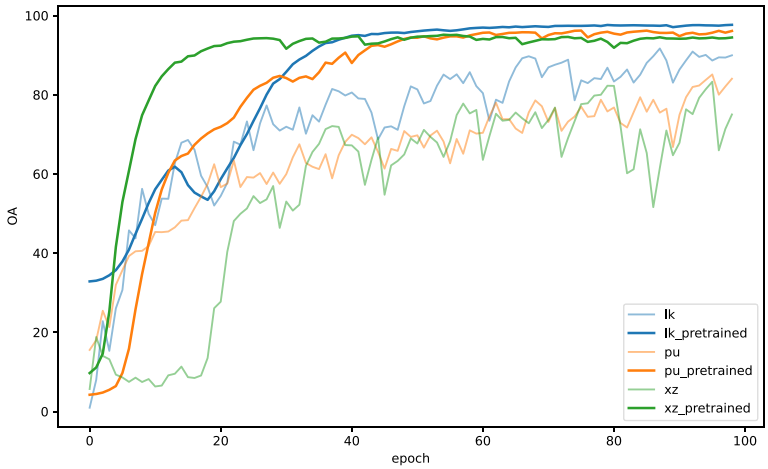
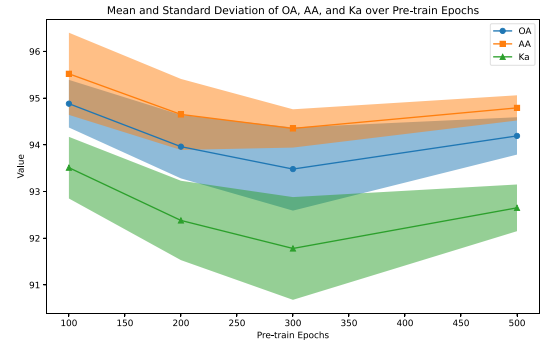
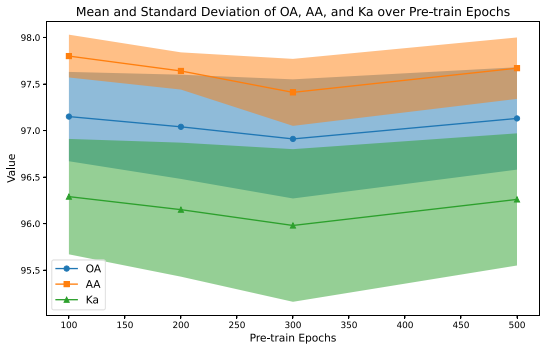


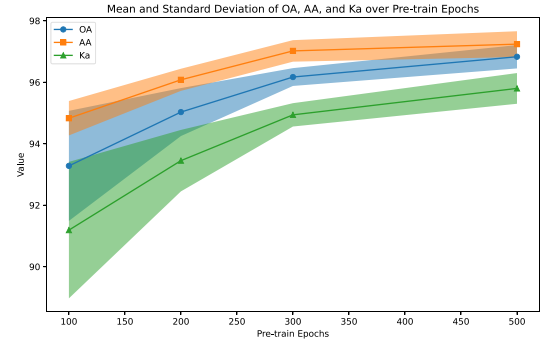
Fig. 11. Performance comparison between training from scratch and fine-tuning after pretrain.



(a)



(b)



(c)

Fig. 12. Different pretrain epoch performance. (a) XZ. (b) LK. (c) PU.

metrics, while the shaded areas indicate the scale of standard deviation. This visualization highlights the robustness of different datasets under masked spectral pretraining, providing profound insights into their behavior. The PU dataset tends to benefit from more extensive pretraining to acquire sufficient latent representations, whereas the XZ dataset exhibits the opposite trend. This suggests variability in data quality and the differing impacts of pretraining across datasets, underscoring the significant research potential of masked spectral tasks as proxy activities.

E. Ablation Studies

In this section, we conduct extensive ablation studies on the proposed pretraining network, specifically including learning rate scaled (LRS), mask ratio (MR), and a number of multihead attention heads. In the first set of experiments, we performed a grid search to identify the optimal pretraining LRS value for each individual dataset. In the second set of experiments, we systematically explored the impacts of different MRs during pretraining on reconstructing spectral features. In the third set of experiments, we evaluated the effects of varying numbers of multihead self-attention heads on masked spectral feature pretraining. Through these ablation experiments, we systematically examine the impact of different design choices on the final performance of the pretrained model. The results provide insights into how each factor affects the self-supervised pretraining process, guiding us to determine the optimal model architecture and training strategy.

1) Learning Rate Scaled: We examine the impact of the LRS, a technique used to adjust the learning rate proportionally with the batch size. This method aims to maintain training stability and improve convergence speed by compensating for changes in gradient updates due to varying batch sizes. Its relationship with the initial learning rate is as follows:

$$\text{learning rate} = \frac{\text{base_lr} \times \text{batch_size}}{\text{LRS}}$$

where base_lr represents the base learning rate and LRS represents the LRS. A larger LRS value indicates a smaller initial learning rate for the network, and vice versa.

To investigate the influence of different learning rate scales on pretraining, we configured the pretraining phase to last 100 epochs, with a batch size of 256. Fine-tuning was conducted using a network consisting of four layers and

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TABLE VII
PERFORMANCE OF XZ DATASET AT DIFFERENT LRS (%)

	64	128	256	512	1024
OA	94.09±1.59	92.58±1.13	92.14±1.96	92.51±1.23	93.75±0.94
AA	94.94±0.94	93.81±0.58	93.90±0.59	93.78±0.65	94.91±1.24
Ka	92.55±1.94	90.69±1.36	90.16±2.35	90.60±1.50	92.12±1.16

TABLE VIII
PERFORMANCE OF LK DATASET AT DIFFERENT LRS (%)

	64	128	256	512	1024
OA	97.15±0.48	97.01±0.61	96.91±0.59	95.78±1.82	96.77±0.46
AA	97.80±0.23	97.30±0.40	97.29±0.31	96.87±0.82	96.88±0.28
Ka	96.29±0.62	96.11±0.79	95.98±0.77	94.54±2.30	95.79±0.59

TABLE IX
PERFORMANCE OF PU DATASET AT DIFFERENT LRS (%)

	64	128	256	512	1024
OA	92.57±1.50	93.08±1.53	93.07±0.51	91.84±0.70	91.06±1.27
AA	94.36±0.39	94.78±0.41	94.94±0.34	93.70±0.54	93.18±0.62
Ka	90.28±1.87	90.93±1.91	90.91±0.62	89.28±0.88	88.29±1.58

two multihead attention heads. The results, as detailed in Tables VII–IX, demonstrate how different LRS affect the pretraining effectiveness, highlighting optimal conditions that enhance model performance during subsequent fine-tuning stages. This study provides insights into the tuning of learning rates to maximize the potential of deep learning architectures in handling complex tasks.

2) Mask Ratio: As a core component of our proposed network, we investigate the pretraining performance under different MRs. As depicted in Table X, the observed trend of the MR's impact on pretraining reflects the distinctive characteristics of our model concerning spectral features masked during training. It is noteworthy that the PU and XZ datasets exhibit larger standard deviations at MR values of 0.75–0.8 compared to other MR values. Despite the requirement for different initial learning rates to expedite convergence across these three datasets, all datasets consistently show a preference for a larger MR within a reasonable range. When studying this MR parameter, we employ the optimal LRS values identified in the previous experiment, while keeping the number of multihead attention heads consistent and pretrain 100 epochs. This approach ensures that our analysis of the MR is isolated from other variables, providing a clearer understanding of its effect on the pretraining process.

3) Heads: The multihead self-attention mechanism, by employing multiple attention heads in parallel, enables the model to capture different aspects of the input data, thereby enhancing the model's representational capacity, computational efficiency, and robustness. Fig. 13 depicts the fine-tuning results of Swin-MSP with 2, 4, 8, and 16 multihead attention heads, respectively. In each SFBT layer, we configure an identical number of attention heads to differentially analyze the impacts arising from variations in head counts across layers. In investigating the influence of attention heads on pretraining, we set pretrain 100 epochs and maintained optimal values for the LRS and MR, as determined in prior experiments. On the LK dataset, the performance of classification decreases as the number of heads increases, and the larger standard deviation indicates that the average classification accuracy is relatively dispersed. In contrast, the XZ and PU datasets do not show sensitivity to the number of heads. The results indicate that the different datasets exhibit contrasting characteristics in pretraining with varying numbers of multihead attention. This phenomenon can be attributed to the significant number of attention heads, which disperses the model's ability to provide fine-grained representations for datasets with dense interclass features. This underscores the importance of considering attention head configurations in pretraining strategies for diverse datasets.

TABLE X
OA OF DIFFERENT DATASETS UNDER DIFFERENT MR (%)

	PU	XZ	LK
0.70	92.75±0.91	92.55±0.70	97.07±0.44
0.75	91.85±1.37	92.82±1.17	97.13±0.67
0.80	92.92±1.09	92.77±1.15	96.95±0.47
0.85	93.07±0.51	93.75±0.94	97.15±0.48

F. Throughput Analysis

In this section, we conducted a comparative analysis of the throughput of different models. Throughput is a crucial performance metric, reflecting a model's ability to process data within a unit of time. The calculation formula is as follows:

$$\text{throughput} = \frac{30 \times \text{batch_size}}{\mathbb{T}}$$

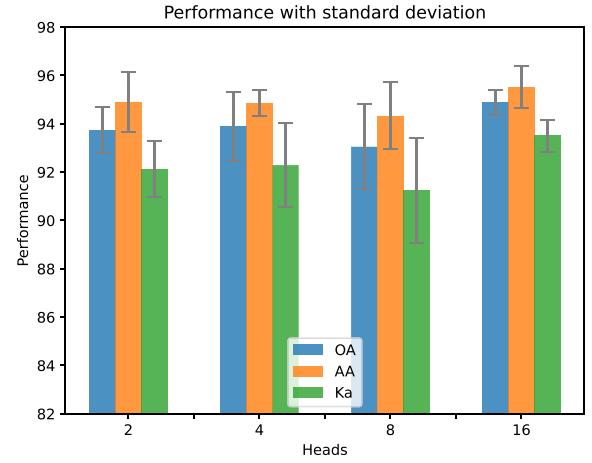
where $\mathbb{T}$ represents the time required to perform 30 inferences with the model given a specified batch size. The results are shown in Table XI. In this experiment, we used consistent hardware, including an Intel[1] Xeon[1] Platinum 8362 with 14 cores and an RTX 3090. We set the batch size to 320 for quantitative analysis. In addition, we utilized GPU warm-up and CUDA synchronization to maintain the consistency of the results. From the results, it can be seen that the proposed network achieves a good balance between classification accuracy and model performance.

G. Discussion

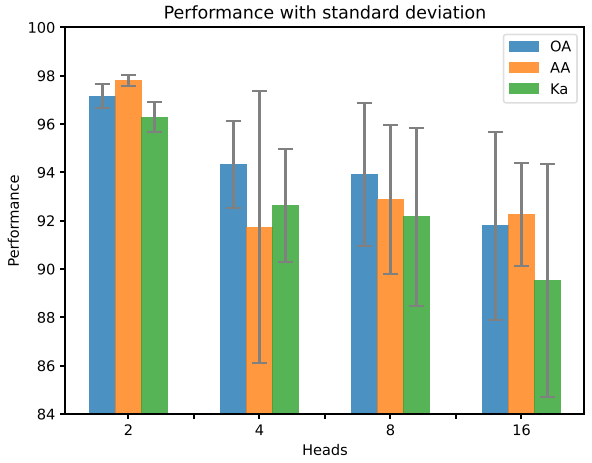
This section systematically explores the application of shift window masking for spectral pretraining on hyperspectral data. After undergoing unsupervised pretraining, the model acquires latent representational capabilities, which are then applied to supervised fine-tuning with a limited number of samples. Extensive experiments demonstrate that the proposed model reliably performs across datasets of varying quality. Moreover, fine-tuning using pretrained weights and biases achieves higher performance and faster convergence in classification tasks. Consistent comparison results with state-of-the-art (SOTA) networks further substantiate this.

[1]Registered trademark.

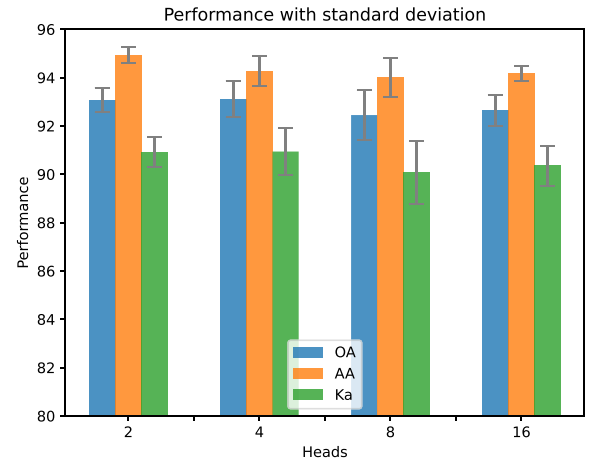
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(a)



(b)



(c)

Fig. 13. Different heads performance. (a) XZ. (b) LK. (c) PU.

Due to hardware constraints, we set the model embedding dimension to 96 and the HSI patch size to 5. These parameters warrant further investigation. In addition, for Windows Masking, we employed random masking on fine-grained patches. Exploring different masking strategies in conjunction with HSI features could be another meaningful research direction. Nonetheless, we have demonstrated that using a shifted window hierarchical architecture with masked spectra achieves robust performance in HSI pretraining. This approach could potentially broaden the scope of future research in the field of HSI classification.

TABLE XI
THROUGHPUT ANALYSIS (USING THE PU DATASET AS AN EXAMPLE)

network	throughput	percent	parameters
1DCNN	300.19K	3159.96%	5K
2DCNN	5.93K	62.40%	10158K
3DCNN	3.55K	37.42%	1068K
SF(pixel)	42.56K	448.02%	155K
SF(patch)	42.07K	442.87%	177K
MAEST(pixel)	42.87K	451.21%	162K
MAEST(patch)	42.09K	443.09%	184K
morphFormer	9.02K	94.94%	223K
SwinMSP	9.50K	100.00%	5045K

V. CONCLUSION

In this study, we developed a novel pretraining model called the Swin-MSP model. We efficiently mapped HSIs into uniaxial vectors and then applied nonoverlapping logical windows for self-attention computation. Utilizing uniaxial shifted windows and a hierarchical architecture, we gradually captured global spatial–spectral information from local contexts. We implemented this architecture and conducted extensive masked spectral unsupervised pretraining on HSIs, followed by transferring the model to downstream supervised classification tasks with limited samples. We compared the classification performance with various SOTA networks and conducted thorough ablation studies. The strong results across multiple benchmarks demonstrate that the proposed model is highly suitable for hyperspectral datasets of varying sizes. We believe these findings provide new perspectives on pretraining strategies and model architectures for HSI classification tasks. Further research is needed to explore the limitations and full potential of pretrained representations to advance HSI analysis across different problem settings.

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